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Morphological Variation of Major Human Populations Based on Nonmetric Dental Traits

Tsunehiko Hanihara*

Department of Anatomy and Biological Anthropology, Saga Medical School, Saga 849-8501, Japan

KEY WORDS inter-regional variation; intra-regional variation; subsaharan Africans; modern human origins; patterns of modern human diversity

ABSTRACT The patterns of inter- and intra-regional variation among 12 major geographical groups from around the world were investigated based on 15 nonmetric dental traits. The R-matrix method was applied using a pooled within-group variance-covariance matrix estimated with the maximum likelihood method (tetrachoric correlation matrix) and the threshold value for each trait estimated by univariate probit analysis. Using average heritability rates that range from 0.40 to 1.00, the inter-regional variation represented by *Fst* falls between 7.19% and 16.23% of the total variance. This range of variation is compatible with those obtained by genetic, craniometric, and odontometric data. Subsaharan Africans show the largest intra-regional diversity among the groups compared. The degree of intra-regional variation shows, moreover, rough clinalities from subsaharan Africa to peripheral regions. The relationship between regional variation and geographic distance from subsaharan Africa supports serial bottlenecks and the founder effect of ancient populations originating in Africa. The variation of East/Northeast Asians is relatively large, suggesting a complex population history such as possible earlier divergence and multiple migrations from outside sources. The present findings are in agreement with both the recent African model for the origin of anatomically modern humans and the current scenario for human migration history suggested by genetic analyses. Am J Phys Anthropol 136:169–182, 2008. © 2008 Wiley-Liss, Inc.

Over the last two decades, most genetic and some morphological evidence have argued in favor of a recent single origin of anatomically modern humans in subsaharan Africa (Cann et al., 1987; Cavalli-Sforza et al., 1988; Stringer and Andrews, 1988; Stoneking and Cann, 1989; Vigilant et al., 1991; Stringer, 1992; Nei and Roychoudhury, 1993; Lahr, 1996; Irish, 1997, 1998; Irish and Guatelli-Steinberg, 2003; Bräuer et al., 2004; Hanihara et al., 2003; Grine et al., 2007; and many others). Until the early 1990s, a focus of the debate was whether subsaharan Africans were the most widely divergent group in a worldwide biological context (Bowcock et al., 1991, 1994; Vigilant et al., 1991; Nei, 1993, 1996; Ruiz-Linares et al., 1993; Cavalli-Sforza et al., 1994). Although many tree diagrams based on genetic data demonstrate that the first-order split is between subsaharan Africans and other regional populations, the results based on analyses of phenotypic data do not necessarily show such distinctiveness of subsaharan Africans (Howells, 1989, 1995; Scott and Turner, 1997; Brace et al., 2001).

In addition to the early important insights about modern human variants (Lewontin, 1972; Latter, 1980; Nei and Roychoudhury, 1982; Ryman et al., 1983), the study of human variation within and among geographic regions has become increasingly important for understanding patterns of modern human diversity and demographic history (Excoffier et al., 1992; Dean et al., 1994; Barbujani et al., 1997; Stoneking et al., 1997; Mountain, 1998; Seielstad et al., 1998; Relethford and Jorde, 1999; Jorde et al., 2000; Brown and Armelagos, 2001; Relethford, 2002; Rosenberg et al., 2002). More recently, several lines of evidence for gradients of genetic and phenotypic diversity within and among continents address the process of the occupation of the present range, drift, and bottleneck effects, and possible natural selection in modern human evolution (Harpending and Rogers, 2000; Ayub et al., 2003; Relethford, 2004; Manica et al., 2005, 2007; Prugnolle et al., 2005; Ramachandran et al., 2005; Ray et al., 2005;Liu et al., 2006; Serre and Pääbo, 2007).

Similar to tree estimation, global patterns of variants at the genetic level have often been contrasted with those measured by phenotypic characters (Nei and Roychoudhury, 1982; Howells, 1989, 1995; Scott and Turner, 1997; Hennessy and Stringer, 2002). However, direct comparison of the variants obtained from quantitative phenotypic and molecular genetic data was not attempted until Relethford and his colleagues developed a multivariate pooled variance approach (R-matrix method) based on an equal and additive effects model (Relethford and Blangero, 1990; Relethford, 1994, 1996,

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*Correspondence to: Tsunehiko Hanihara, Department of Anatomy and Biological Anthropology, Saga Medical School, 5-1-1 Nabeshima, Saga 849-8501, Japan.
E-mail: hanihara@cc.saga-u.ac.jp

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TABLE 1. Standards for the 15 nonmetric dental traits

Trait observed	Abbreviation	Criteria
1. Shoveling (UI1)	SH	The depth of Lingual fossa deeper than 0.5 mm is scored as present. This classification corresponds to the categories of semi, moderate, and strong shovel employed by Hanihara et al. (1970), and grade 3–7 of the Arizona State University Dental Anthropology System (ASU system) (Turner et al., 1991; Scott and Turner, 1997).
2. Double shoveling (UI1)	DSH	Following the ASU system, the grade 3–6 is scored as present.
3. Shoveling (UI2)	SH	The teeth classified as the ASU system's grade 3–7 are scored as present.
4, 5. Premolar accessory cusp (UP1, UP2)	PAC	This trait is recorded following the ASU system.
6. Carabelli's cusp (UM1)	CC	Based on Dahlberg's (1963) criteria, presence (Dahlberg's d–g) is distinguished from absence (Dahlberg's a–c). The grade a–c is equivalent to 0–2 of the ASU system.
7. Hypocone (UM2)	HYC	Presence corresponds to 4- and 4+ of Plaque P9, Dahlberg (1949) and grade 3–5 of the ASU system.
8. Central ridge (LP1)	CR	Deutero-proto relationship. The scoring procedure followed is one by Higa et al. (2003).
9, 10. Sixth and seventh cusp (LM1)	C6, C7	The counting method of the two traits is the same as that of the ASU system. Scored cusp as present regardless of the size.
11. Deflecting wrinkle (LM1)	DW	Following Weidenreich (1937), if the medial ridge of metaconid deflected distally, the trait is scored as present. Presence corresponds to grade 2–3 of the ASU system.
12. Protostylid (LM1)	PRS	The ASU system's 2–7 were scored as present.
13. Distal trigonid crest (LM1)	DTC	Following the original description by Weidenreich (1937), the crest connecting the tip of the metaconid with the distal accessory ridge of the protoconid without interruption is scored as present. This method is the same as that of the ASU system.
14. Hypoconulid (LM2)	HYCD	The counting method is the same as that of the ASU system.
15. Sixth cusp (LM2)	C6	Follows the ASU system

2001; Relethford and Harpending, 1994, 1995). Using the same approach used for genetic traits, several studies demonstrated the fit of the global patterns of human phenotypic variation to the evolutionary model constructed from genetic data (Relethford, 2002, 2004; Roseman, 2004; Hanihara and Ishida, 2005; Manica et al., 2007). However, the methods used in these studies essentially require morphometric data.

Nonmetric dental (and cranial) traits, that are generally believed to be under moderate to high genetic control and selectively more neutral than metric traits, have proven useful for investigating microevolutionary mechanisms at the infraspecific level in various animals including hominids (Ossenberg, 1986; Hauser and De Stefano, 1989; Scott and Turner, 1997; Hanihara et al., 2003). Moreover, the intertrait correlations of nonmetric traits are relatively low in comparison with those based on dental (and cranial) measurements (Hanihara, 1976; Scott and Turner, 1997; Dodo et al., 1998). These attributes suggest that nonmetric data may better fit the assumptions required in the R-matrix method than quantitative morphological data. Although Pilbrow (2006) applied modified R-matrix statistics to nonmetric dental data for various species of gibbons, this method has not been used to a similar level of success with nonmetric data (Leigh and Konigsberg, 1996; Leigh et al., 2004; Pilbrow, 2006).

The purpose of this article is to extend the R-matrix method to nonmetric dental data for estimating the apportionment of global human morphological diversity as well as diversity within geographic regions on a world-wide scale. Moreover, this study examines the fit of the isolation-by-distance model (Relethford, 2004) to the global pattern of phenotypic variation using simple regression analysis. These approaches provide estimation of the insight into joint effects of the determinants of phenotypic variation such as bottlenecks, isolation, natural selection, or admixture (Ramachandran et al., 2005).

MATERIALS AND METHODS

The 12 geographic samples used in this study and brief descriptions of each are given in the Appendix. Detailed information including country of origin, tribal affiliation, and other regional attributes for these series are presented elsewhere (Hanihara, 2000; Hanihara and Ishida, 2001). In this study, samples older than 2,000 years were excluded to avoid any bias in the quality of the materials, except for the Egyptian and Nubian samples. Irish (2005, 2006) demonstrates that both the Egyptian and Nubian dental series from Predynastic to recent times through the Christian period exhibit relative homogeneity, suggesting overall post-Neolithic diachronic and regional population continuity.

Morphological observations were made on 15 variables in the permanent dentition according to the procedures shown in Table 1. All the data were recorded by the present author to avoid possible interobserver error. The traits recorded are limited to crown characteristics, because the data from the Japanese and Negritos were taken from plaster casts. The 15 traits used in this study include characteristics attributed to the Mongoloid (Hanihara, 1968), Caucasoid (Mayhall et al., 1982), and subsaharan African dental complex (Irish, 1997, 1998), as well as the key crown traits that distinguish sundadonty from sinodonty (Turner, 1990). The phylogenetic significance of the crown traits used here is discussed elsewhere (Hanihara, 1992; Higa et al., 2003).

Although observations were made on both the right and left antimeres for each trait, the individual count method was used; thus, if a trait was present on either or both sides, it was scored as being present (Turner et al., 1991; Hillson, 1996; Hanihara et al., 2003). In addition to the intertrait correlations, the tetrachoric correlation coefficients between every pair of traits in each geographic sample were also calculated. The sexual differences of the frequencies were tested using the conventional χ^2 test and the Fisher's exact probability test.

For estimating both inter- and intra-regional variations in quantitative phenotypic features, Relethford and Blangero's (1990) R-matrix approach has been applied on a global as well as a local scale (Relethford, 1994, 2002; Relethford and Harpending, 1994; Powell and Neves, 1999; Steadman, 2001; González-José et al., 2003, 2005a,b; Roseman and Weaver, 2004; Stojanowski, 2004; Hanihara and Ishida, 2005; Sardi et al., 2005; Schilacii and Stojanowski, 2005; Scherer, 2007; Toma et al., 2007). Hallgrímsson et al. (2004) used nonmetric morphological trait frequencies as analogous to allele frequencies in computing the R-matrix and other genetic parameters without any theoretical justification.

It is well known that the traits called nonmetric (discrete) dental and skeletal variants refer to dichotomous characteristics with a continuous underlying distribution of liability that are assumed to be polygenically determined. The model assumes that the trait liability is normally distributed, and dichotomized by the imposition of a threshold (Hauser and De Stefano, 1989; Konigsberg, 1990; Blangero and Williams-Blangero, 1991, 1993).

In this study, the R-matrix method modified for morphometric data (Relethford, 1994; Relethford and Harpending, 1994) was applied to worldwide human nonmetric dental data using a pooled within-group variance-covariance matrix estimated with both the maximum likelihood method (tetrachoric correlation matrix) and the standard deviation units derived from threshold values for each trait. The thresholds were estimated using a univariate probit analysis (Konigsberg, 1990; Konigsberg et al., 1993). The tetrachoric correlation matrix estimated with the maximum likelihood method is an extension of Relethford and Blangero's (1990) pooled within-group additive genetic variance-covariance matrix to cover nonmetric traits caused by thresholds on polygenically determined liabilities (Blangero and Williams-Blangero, 1991; Konigsberg et al., 1993).

The application of the R-matrix method to morphological data requires an estimate of average heritability for the phenotypic traits. Previous studies have reported the heritability estimates of tooth crown traits based on family and twin studies (Biggerstaff, 1970, 1973; Berry, 1978; Hillson, 1996; Scott and Turner, 1997). Mizoguchi (1977) reported the heritability of 14 crown traits for Japanese. Eleven out of the 14 traits are the same as those used here, and the mean heritability rate estimated for the 11 traits was calculated to be 0.52 (the range from 0.22 to 0.75). If heritability is a population-specific value, however, the estimates cannot be applied to other samples and groups, or across the board. Several researchers studying twin series from different populations arrived at similar heritability estimates for some traits (Hanihara et al., 1975; Blanco and Chakraborty, 1977; Mizoguchi, 1977; Scott and Potter, 1984; Townsend et al., 1992). Surveying several studies (Sofaer et al., 1972; Mizoguchi, 1977; Scott and Potter, 1984; Townsend et al., 1992), it is clear that most of the estimates of heritability rates for crown traits fall in the 0.40–0.80 range.

RESULTS

The incidence of the 15 crown traits in the male and female samples and the results of the χ^2 as well as the Fisher's exact probability test for differences between sexes are given in Table 2. The frequency distributions by sex were not significantly different for most of the samples. Table 3 provides the tetrachoric correlation coefficients calculated within each sample and averaged over all samples weighted by sample sizes. The results suggest that the intertrait correlations of the 15 nonmetric traits including those between single morphological traits (e.g., interstitial tubercle and sixth cusp) and variants of the same trait (shoveling) recorded on different teeth are sufficiently low in comparison with those based on the dental and cranial measurements (e.g., Constandse-Westermann, 1972; Hanihara, 1976; Hillson, 1996; Dodo et al., 1998). For these reasons, I have selected to use the sex-combined frequencies of the 15 crown traits.

Assuming that the samples investigated have the same effective population size, inter-regional variation among 12 geographic groups was estimated by *Fst* values. The minimum *Fst* and estimated *Fst,* using average heritability values that range from 0.40 to 0.90 for the nonmetric crown traits, are given in Table 4. The results show that nonmetric dental variation across the worldwide regions is limited.

Table 5 gives the results of Relethford and Blangero's (1990) method for estimating intra-regional variations under the assumption that the effective population size is equal and the average heritability rate of the dental traits is 0.5. This heritability rate may not be an overestimated value based on the results of Turner (1989) and Hillson (1996). The levels of observed variance in each geographic region given in Table 5 are visually expressed in Figure 1. Deeper gray values represent larger variance. The subsaharan African sample shows the largest within-group variation, followed closely by the West Asian sample. The average phenotypic variations of South Asian and East/Northeast Asian samples together with the two Oceanian samples, Melanesian and Pacific samples, are greater than those of the North African, European, and Southeast Asian samples. The Australian and the New World samples show relatively low intra-regional variation.

In the Relethford and Blangero's (1990) R-matrix method, equal weighting of population size is used as a null hypothesis such that the long-term effective population size of different geographic region is the same or reveals constant effect of long-range gene flow between regions (especially in the case of peripheral or local populations). When the null-hypothesis can be refuted, the residual variance, the difference between observed and expected within-group variances, is statistically significant (not equal to zero) (Relethford and Harpending, 1994). The distribution of the residual variance divided by its standard error is approximated by a *t*-statistic with $t - 1$ degrees of freedom (t; the number of traits). The standard errors of the residual variance are estimated by Jack-knife method (Miller, 1974; Relethford and Harpending, 1994). The results of the significance test are given in Table 5. The European, Southeast Asian, and the two New World samples show signifi-

TABLE 2. Frequencies of the 15 crown traits and the sexual dimorphism tested by χ^2 and Fisher's exact probability test

Sample name	Sex	SH (UI1)		DSH (UI1)		SH (UI2)		PAC (UP1)		PAC (UP2)		CC (UM1)		HYC (UM2)		CR (LP1)		C6 (LM1)		C7 (LM1)		DW (LM1)		PRS (LM1)		DTC (LM1)		HYCD (LM2)		C6 (LM2)	
	Male/ Female	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N
East/Northeast	M	95.9	98	20.8	96	92.1	114	28.7	164*	8.7	172	9.4	286	87.1	264	61.9	134**	28.3	173	10.0	220	38.7	111	7.7	170	1.9	209	32.8	183	7.9	152
Asia	F	100.0	54	25.5	55	93.9	66	41.6	89	9.9	91	8.4	131	82.6	121	81.3	75	27.1	70	5.1	78	35.2	54	14.5	62	2.3	89	31.0	71	7.6	66
Southeast	M	72.8	147*	6.7	135	76.7	172	19.9	341	12.3	326	16.0	656	92.0	634*	80.9	256	25.4	331	8.6	428	29.8	289	8.7	347	2.7	401	45.7	414*	9.3	365
Asia	F	87.0	46	14.7	34	83.9	56	13.5	104	9.2	109	12.9	263	87.4	230	79.2	72	19.4	124	8.0	151	23.8	101	8.3	121	0.7	137	57.5	134	12.6	119
Pacific	M	73.2	112	6.1	49	73.5	162	11.5	157	3.5	143	13.8	457	94.0	451	85.9	99	39.8	241	6.2	292	31.2	215	9.1	253	3.5	285	40.7	290	15.0	260
	F	78.0	127	0.0	28	76.2	164	10.4	77	2.8	71	14.2	373	92.0	298	83.8	37	47.2	216	5.0	241	33.3	198	11.0	218	5.7	227	43.5	232	11.8	228
Melanesia	M	69.2	78	7.7	78*	75.6	123	11.4	184	27.8	187	16.8	471	97.6	459	91.1	135	47.6	208*	12.9	279	52.8	178	3.8	238	7.0	271*	32.5	246	13.0	192
	F	70.8	48	0.0	49	67.1	76	13.3	120	23.9	113	13.5	386	95.5	357	83.5	85	37.4	182	9.0	244	47.3	169	2.3	218	2.9	243	39.6	212	9.1	186
Australia	M	65.7	67	1.3	75	65.8	111	19.5	87**	41.4	87	5.4	168	97.8	278	89.6	67	67.1	79	6.5	138	62.1	66	8.1	99	1.7	115	6.0	167	48.3	89*
	F	72.0	25	0.0	22	60.0	40	0.0	33	26.7	30	5.6	89	99.1	107	93.3	30	54.6	44	6.2	65	70.0	30	8.2	49	0.0	53	10.0	80	28.9	52
North	M	99.5	192	43.2	185	97.1	240	14.6	178	2.5	158	2.8	394	88.0	450*	92.5	187	44.2	163	6.0	282	53.3	90	8.3	192	2.2	274	13.9	288*	28.0	161*
America	F	98.1	160	40.2	137	95.3	193	15.7	147	5.2	136	2.2	358	82.5	366	93.3	149	31.6	152	5.5	237	53.2	126	10.1	189	1.8	229	20.4	235	17.0	165
South	M	97.4	39	37.5	40	88.7	53	15.1	73	7.6	66	1.5	204	90.6	233	85.7	49	29.0	93	6.3	142	39.1	64	8.7	103	1.5	130	18.1	127	15.7	83
America	F	100.0	25	34.8	23	94.1	34	14.9	47	2.4	41	1.2	161	85.8	162	87.1	31	34.3	67	2.3	87	38.9	54	4.4	69	2.5	81	23.6	72	13.8	58
India	M	64.4	59	1.4	71	70.0	80	5.3	132	6.2	113	11.9	270	86.8	319*	74.4	86	10.5	124	11.3	186	19.6	92	6.6	151	1.2	172	56.3	160*	4.2	119
	F	47.4	19	0.0	24	63.3	30	3.3	60	7.9	63	6.7	119	78.3	129	80.9	47	10.9	55	7.0	86	22.9	48	1.5	68	1.2	84	72.0	75	0.0	68
West Asia	M	28.6	28	0.0	30	51.3	39	6.7	75	5.3	76	35.0	143	79.7	172	54.6	33	12.2	41	16.1	56	9.7	31	0.0	49	0.0	54	64.0	50	2.3	43
	F	57.1	7	0.0	7	50.0	10	0.0	33	3.0	33	27.1	85	72.0	75	80.0	10	10.0	10	14.3	14	9.1	11	0.0	15	0.0	15	76.9	13	0.0	13
Europe	M	32.8	241	1.8	273	43.1	281	4.2	406	5.4	389	24.1	524	77.0	655	74.7	359	5.7	383	8.1	459	16.8	358*	2.2	402	0.7	447	73.9	422	1.2	403
	F	34.6	104	0.9	106	46.0	126	3.4	176	3.1	163	19.2	214	72.2	241	72.0	150	6.5	154	9.4	171	9.2	141	4.4	158	1.2	166	78.9	166	1.2	161
North	M	19.1	21	0.0	24	41.9	31	2.5	79	2.6	77	26.8	142	90.8	228*	66.7	57	8.8	80	12.5	112	16.1	56	1.3	75	2.8	106	60.2	108	1.1	94
Africa	F	33.3	6	0.0	5	16.7	12	0.0	48	0.0	52	18.8	144	82.8	163	80.0	20	4.3	47	17.5	63	32.4	34	2.1	47	0.0	58	59.4	69	1.5	68
Subsaharan	M	53.1	98	0.9	110	56.8	139	4.6	346	9.8	338*	18.6	570	91.4	614	70.9	268	23.8	269	34.6	350	46.1	219	2.1	292	0.6	332	29.3	348*	14.4	270
Africa	F	41.7	24	0.0	27	47.7	44	5.0	140	3.7	136	13.8	261	87.6	265	65.1	86	17.1	88	32.5	120	41.3	75	0.0	107	1.7	117	41.6	113	12.9	93

* Significant at 5% level.
** Significant at 1% level.

TABLE 3. Tetrachoric correlation coefficients calculated within each sample and pooled using sample size for each trait pair

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Shoveling (UI1)	–														
2. Double shoveling (UI1)	0.10300	–													
3. Shoveling (UI2)	0.37134	0.16753	–												
4. Accessory cusp (UP1)	0.07895	0.03966	0.00130	–											
5. Accessory cusp (UP2)	0.11417	0.02861	0.08326	0.06111	–										
6. Carabelli's cusp (UM1)	0.03926	0.04853	−0.09142	0.08083	0.14300	–									
7. Hypocone (UM2)	0.06319	0.12181	0.00716	0.03699	0.06686	0.13744	–								
8. Central ridge (LP1)	0.07075	0.11462	0.05419	0.03990	−0.16871	0.11137	0.02144	–							
9. Sixth cusp (LM1)	−0.15455	0.04248	0.11009	0.07155	0.09738	0.02228	0.03522	0.11088	–						
10. Seventh cusp (LM1)	−0.06945	−0.19338	0.03564	0.50615	0.04145	0.06913	0.06878	0.20777	0.04245	–					
11. Deflecting wrinkle (LM1)	0.07137	0.08236	0.04239	−0.11623	0.14406	0.11316	−0.01213	−0.00180	0.14049	0.06982	–				
12. Protostylid (LM1)	0.11415	0.14727	0.17779	−0.05213	0.05659	0.28823	−0.11114	0.15356	0.21183	0.01807	0.07393	–			
13. Distal trigonid crest (LM1)	0.41799	−0.12126	0.10192	0.01664	0.23430	0.14061	0.03711	0.00052	0.02173	−0.31903	0.14449	−0.01392	–		
14. Hypoconulid (LM2)	−0.10022	0.06936	0.04289	−0.05299	−0.10948	0.07610	0.09081	0.18214	0.33764	0.26141	−0.35812	0.38033	0.11645	–	
15. Sixth cusp (LM2)	0.09108	0.24508	−0.11017	0.15366	0.16368	0.19995	−0.21301	0.17664	0.24858	−0.15659	0.29382	−0.00825	0.08482	−0.35980	–

TABLE 4. Minimum and estimated Fst values

Heritability	Estimated *Fst*	S.E.
0.40	0.1623	0.0136
0.50	0.1342	0.0130
0.60	0.1144	0.0124
0.70	0.0997	0.0119
0.80	0.0883	0.0114
0.90	0.0793	0.0110
1.00 (minimum)	0.0719	0.0106

cantly less average phenotypic variation than expected variation, suggesting smaller population size, genetic drift, and/or less inter-regional migration rates (Relethford and Harpending, 1994).

Following the largest within-group phenotypic diversity in subsaharan Africa, the relationship between the degree of intra-regional variation and geographic distance is examined by applying simple regression analysis. The geographic distances between subsaharan Africa and other regions used in this study are reported by Manica et al. (2007). The Melanesian and Pacific samples are excluded in the analysis here because there is little contribution to the gene pool of the Eurasian and New World populations, as discussed below. Figure 2 shows the result of the regression analysis. The linear regression for the within-group variation on geographic distance produces a value of $Y = -2\text{E-}06\ X + 0.155$, and the percentage of variance explained by geographic distance is $R^2 = 0.6747$ ($P < 0.001$). The negative linear correlation is significant, with the populations most geographically distant from subsaharan Africa characterized by lower phenotypic variability.

DISCUSSION

In morphological studies, researchers are faced with the need to explain the evolutionary processes by which morphological variation likely developed (Lahr, 1994, 1995, 1996; Relethford and Harpending, 1994; Lahr and Wright, 1996). Relethford (1994, 2002, 2004) suggests that inter-regionally differing selection pressures have played only a limited role in producing an overall pattern of craniofacial morphological diversity. However, it is well known that differences in metric dental and cranial characteristics among regions are in part the product of inter-regionally differing selection pressures (Beals, 1972; Guglielmino-Matessi et al., 1979; Carey and Steegmann, 1981; Beals et al., 1983; Brace et al., 1991; Roseman, 2004; Harvati and Weaver, 2006). Roseman and Weaver (2004) point out that some craniometric features display higher levels of interpopulation differentiation than randomly selected genetic loci by applying both the R-matrix method and principal component analysis.

In this study, the estimated values of *Fst* fall within a range of 0.163–0.079 (minimum *Fst* = 0.072), and intra-regional variation is highest in subsaharan Africa. These results are consistent with previous results showing low levels of inter-regional variation and a higher level of within-regional diversity in subsaharan Africa than in other geographic regions reported in studies that use genetic data (Vigilant et al., 1991; Stoneking, 1993;

TABLE 5. Observed, expected, residual variances, and average distances from Africa

Region	Observed variance	Expected variance	Residual variance	S.E.	Distance from Africa (km)
East/Northeast Asia	0.1453	0.1490	−0.0038	0.0058	13,893
Southeast Asia	0.1358	0.1510	−0.0152**	0.0043	14,529
Pacific	0.1401	0.1443	−0.0042	0.0065	–
Melanesia	0.1469	0.1460	0.0009	0.0090	–
Australia	0.0981	0.1145	−0.0164	0.0092	20,592
North America	0.1177	0.1385	−0.0208***	0.0030	21,969
South America	0.0813	0.1358	−0.0546***	0.0091	31,187
India	0.1424	0.1404	0.0020	0.0146	9,971
West Asia	0.1516	0.1418	0.0098	0.0058	5,765
Europe	0.1218	0.1326	−0.0108*	0.0045	8,441
North Africa	0.1238	0.1424	−0.0180	0.0137	4,607
Subsaharan Africa	0.1570	0.1529	0.0041	0.0101	–

Absolute values of residual variance are significantly larger than zero at 5% level (*), 1% level (**), and 0.1% level (***).



Fig. 1. Visual expression of the magnitude of intra-regional variation in 12 regions given in Table 5. The gray density is adjusted in gray scale tones roughly proportional to the values of intraregional variation (Table 5). Deeper gray values represent larger intraregional variation. Subsaharan Africans shows greater average phenotypic variation than any other regions of the world. On the other hand, Australians, Native Americans, and to a lesser extent, Europeans show less average phenotypic variation.

Jorde et al., 1995; Armour et al., 1996; Tishkoff et al., 1996; Brown and Armelagos, 2001).

Taking the numerical scale and worldwide synthesis of global geographic regions used in this study into account, the present results suggest that nonmetric dental variation, when considered as a whole, has not been subject to strong selection pressures during the recent past, and varies across regions in a manner generally matching the genetic expectations.

The levels of variation among and within major geographic regions have implications for the ongoing debate over modern human origins (Relethford, 1994; Mountain, 1998). Relatively low among-group variation and high subsaharan African diversity have been interpreted as indicative of a relatively recent divergence of modern humans (Jorde et al., 2000; Brown and Armelagos, 2001), and suggests that more time had elapsed in subsaharan Africa since a common origin than in any other region (Stoneking and Cann, 1989; Vigilant et al., 1991; Stoneking, 1993; Jorde et al., 1995; Armour et al., 1996; Tishkoff et al., 1996). The multiregional model emphasizes that low inter-regional variation reflects continued gene flow between geographic regions from a much earlier time (Wolpoff et al., 1984; Wolpoff, 1992). Some genetic studies have, however, criticized the continued gene flow emphasized by the proponents of the multiregional model (Rouhani, 1989; Harpending et al., 1993; Stoneking, 1993). Several researchers have argued that higher diversity within subsaharan Africa is expected under the hypothesis that the average long-term effective population size was large relative to other regions (Relethford and Harpending, 1994; Jorde et al., 1997; Stoneking et al., 1997; Mountain, 1998; Relethford and Jorde, 1999).

The present results shown in Table 5 do not necessarily suggest that the long-term average population size in subsaharan Africa was larger than any other region.

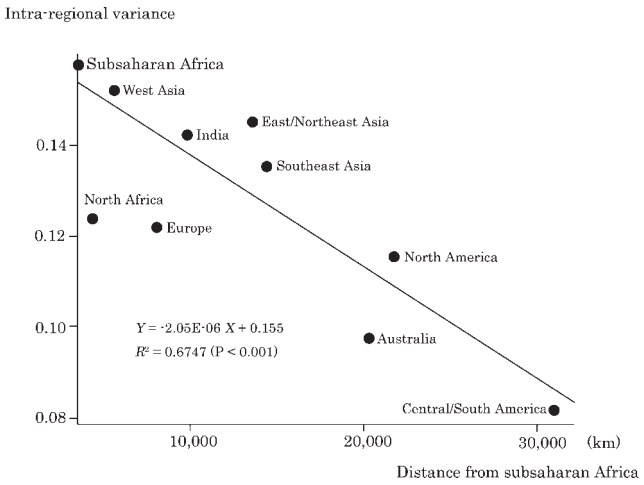


Fig. 2. Relationships between intra-regional variation given in Table 5 and distance from the putative African origin (represented by centroid of locations of the subsaharan African samples).

However, some peripheral groups such as Europe, Southeast Asia, and the New World show significantly less average phenotypic variation, suggesting smaller effective population sizes or lesser long-range gene flow than other regional groups (Relethford, 1994).

The works by González-José et al. (2005b) and Sardi et al. (2005) using *Fst* statistics in cranial comparisons demonstrate high variation in South America when compared with other geographical regions. However, these studies include both recent, prehistoric, and Palaeoindian samples to measure gene flow from external sources, the antiquity of the peopling of South America, the effects of genetic drift, and other tests. The presence or absence of extrogenous contribution such as multiple waves from different sources and possible outliers will affect the results and interpretations regarding levels of within-group variation (Lahr, 1995).

The degree of intra-regional variation shown here has two directional clinalities: one is from subsaharan Africa to Southeast Asia and finally to Australia through West and South Asia; the other is from subsaharan Africa to Europe through West Asia or North Africa. In addition to these two possible clines, the levels of variation decrease from north to south in the New World (see Fig. 1). However, before looking at the possible implications of such gradients, it is necessary to consider the peripheral regions with large within-group variation.

In Table 5, the Melanesian and Pacific samples (Micronesians and Polynesians) show relatively large intra-regional variance. Such large variation may be based at least in part on the differential patterns of gene flow from an outside source, geographical isolation, and genetic drift in each island of the Pacific Islands (Hurles et al., 1998; Richards et al., 1998; Gray and Jordan, 2000; Lum and Cann, 2000; Matisoo-Smith and Robins, 2004). Except possibly for Papua New Guineans (e.g., Tommaseo-Ponzetta et al., 2002; Cann and Lum, 2004; Merriwether et al., 2005), the Pacific islanders represent newly diverged peripheral populations, and contribute little to the global gene pool of living human populations.

On the other hand, the relatively large intra-regional variation of the East/Northeast Asian region suggests long-term population history and/or long-range gene flow from outside sources (Relethford and Harpending, 1994). The classic view held that the human occupation of East/Northeast Asia resulted from the expansion of Southeast Asian people (Turner, 1989, 1990; Li and Bing, 2000). Recently, multiple migrations to the East/Northeast Asian region, such as from somewhere around the western half of Eurasia via the northern Siberian, Central Asian, and South Siberian routes, are inferred by many genetic studies (Underhill et al., 2001; Wells, 2002; Uinuk-Ool et al., 2003; Hong et al., 2005; Hill et al., 2007). The present results may support the complex population history of East/Northeast Asians rather than the simple Southeast Asian model of East/Northeast Asian origin.

Regardless of different population structures in each geographic region, the gradients of the diversity presented herein indicate that geographic distance from subsaharan Africa is a significant and primary determinant of nonmetric dental variation observed on the vast Eurasian, Australian, and New World regions as shown in Figure 2. It has often been predicted that genetic differentiation between populations increases essentially with geographic distance (Relethford, 2004; Ramachandran et al., 2005). Moreover, the rapid expansion out of Africa would imply a progressive loss of genetic diversity through a series of bottlenecks (Prugnolle et al., 2005; Liu et al., 2006; Manica et al., 2007; Serre and Pääbo, 2007). If this is true, as seems likely, the geographic distance from subsaharan Africa along likely colonization routes is one of the strongly supported predictors for not only genetic but also phenotypic diversity of modern human populations. The pattern of decrease in dental variation with distance from East Africa is more or less smooth and provides no suggestion for major discontinuities that could be interpreted as evidence for a second or multiple origin(s) of modern human populations. Therefore, the globally distributed populations can be explained by an expansion from subsaharan Africa.

The present findings are suggestive of a subsaharan African origin for anatomically modern humans, or at least a primary role for subsaharan Africa regarding the evolution of modern humans (Underhill et al., 2000; Yuehai et al., 2001; Ayub et al., 2003; Cruciani et al., 2004; Luis et al., 2004; Mellars, 2006; Manica et al., 2007). Because long-term migration between all pairs of regions is not taken into consideration, the present results do not necessarily rule out the multiregional model (Wolpoff et al., 1984; Wolpoff, 1992) However, the results here suggest that the distribution of human variation of dental traits mirrors that of previous genetic work on human diversity that support the single African origin model.

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APPENDIX . Materials used and brief information (parenthesis, number of individuals examined)

Large geographic groups	Local groups	Brief information
1. East/Northeast Asia	Buryats (36)	Recent Buryats from Kiachta, Troiskosavsk (NMNH, AMNH, MH)
	Amur Basin (30)	Nivkhs, Ulch, Nanai and other people, including a few specimens of Yakuts and Negidals (NHM, MH, AMNH)
	Mongol (224)	Recent Mongolians from Ulan Bator (AMNH, NMNH, MH)
	Japan (71)	Recent Japanese from Tokyo and Sapporo, collected by Kazuro Hanihara in 1960s (plaster cast) (UT)
	Northern China (66)	Recent Northeastern Chinese from Heilongjiang and Jirin Provinces (region formerly referred to as Manchuria) (UT, KU, AMNH)
	Southern China (63)	Recent Chinese from south of Cheng Kiang River (NHM, MH, AMNH)
	Tibet (77)	Recent Tibetans from Sikang and Kangting Provinces (NHM, UC, AMNH)
2. Southeast Asia	Vietnam (94)	Recent Vietnamese, mainly from Hanoi, including a few samples from Laos and Cambodia (MH)
	Thailand (61)	Recent Thai, mainly from Bangkok (AMNH, NHM, MH)
	Myanmar (130)	From Arracan Hills, Rangoon, etc. (NHM, CU)
	Malay (68)	Recent Malayans from Malay Peninsula, mainly from Singapore (AMNH, NHM, CU, FM)
	Andaman/Nicobar Islands (109)	Recent inhabitants of the Andaman Island chain, Jarawa, and of the Nicobar Islands (NHM, UC, MH)
	Sumatra/Java (92)	Recent Sumatrans, including Nias, Banka, and other Islands, as well as Javanese, mainly from Jakarta (NHM, UC, MH, AMNH)
	Borneo (111)	Native inhabitants of Borneo Island, mainly the Land Dayaks (NHM, UC, MH, AMNH)
	Celebes/Molucca (50)	Recent inhabitants of the Celebes and Molucca Islands (NHM, UC)
	Lesser Sunda (28)	Recent inhabitants of the Lesser Sunda Islands: Timor, Bali, Sumbawa, and Flores Islands (NHM, UC, MH, AMNH)
	Philippines (173)	Recent Filipinos, Tagalog, Bilan, Bisaya, Igorot, Ifugao and other tribes; mainly from Luzon and Mindanao Islands (MHM, UC, MH, FM)
	Negritos (55)	Recent Philippine Negritos; Aeta and Agta from Luzon (UT, NHM, MH, AMNH, NMNH)
3. Pacific	Mariana (193)	Recent inhabitants from Guam, Saipan, Tinian (BM, MH)
	Caroline (40)	Recent inhabitants from Ponape, Yap, Palau, Mortlock, Fais (UT, MH, AMNH)
	Tonga/Samoa (30)	Recent inhabitants of Tonga and Samoa Islands (BM, AMNH, NHM)
	Society (97)	Recent Society Islanders, mainly from Tahiti (NHM, MH)
	Cook (33)	Recent inhabitants from Rarotonga, Mangaia (KU, NHM)
	Marquesas (200)	Recent inhabitants from Tahuata, Hatuatua, Nuku Hiva, Ua Huka (NHM, MH, AMNH, BM, FM)
	Hawaii (496)	Recent indigenous people from Oahu, Lanai, Kauai, Molokai, Maui, Hawaii (BM, NHM, UC)
	Easter (79)	Mainly from Rapa Nui Island, including a few specimens from Motu Iti, Motu Nui Islands (NHM, UC, MH, FM)
	Maori (121)	Recent Native people of New Zealand (AM, US, SAM, NHM, UC, AMNH, NMNH)
	Moriori (58)	Recent Native people of Chatham Islands (AM, US, SAM, NHM, UC, AMNH)
4. Melanesia	New Guinea (317)	Purari River Delta, Fly River Delta, Sepik River Delta, etc. (AM, US, SAM, NHM, UC, FM)
	Torres Strait (104)	Mer, Hammond, Thursday, Darnley, Prince of Wales Islands (NHM, UC, MH)
	New Britain (215)	Gazelle Peninsula, Ralum, Matupi, Branche Bay (AM, US SAM, NHM, UC, AMNH, NMNH)
	New Ireland (74)	Recent Native people of New Ireland (AM, US, SAM, NHM, UC, MH, AMNH, FM)
	Solomon (100)	Quahguahroo, North Coast of New Georgia, including a few samples from Santa Cruz Islands (Trevanion Island, etc.) (AM, US, SAM, NHM, UC, MH, AMNH, FM)
	Vanuatu (144)	Malakula, Ambrym, Mallicolo, Tanna Islands (AM, US, SAM, NHM, UC, AMNH, FM)
	New Caledonia (44)	Recent inhabitants from Banks and Royalty Islands (AM, US, SAM, NHM, UC, AMNH, FM)
	Fiji (66)	Recent inhabitants from Viti Levu, Vanua Levu, and Kandavu Islands (AM, US, SAM, NHM, UC, BM, AMNH)

(*continued*)

APPENDIX . (Continued)

Large geographic groups	Local groups	Brief information
5. Australia	Northern Territory/ Queensland (48)	Recent Native Australians from Northern Territory and Queensland (AMNH, NMNH, UC)
	New South Wales/Victoria	Recent Australians from the coastal region of NSW and Victoria (AM, NHM, UC, MH, AMNH)
	Murray River (79)	Recent specimens from Roonka site, and several individuals from Murray river basin (SAM, NHM, UC)
	South Australia (138)	Recent Australians living near Adelaide (SAM, NHM, UC, AMNH, NMNH, MH)
	Tasmania (19)	Recent Tasmanians (AMNH, NHM, UC, MH, FM)
	Western Australia (80)	Recent Australians from Western Australia (NHM, UC, MH, AMNH)
6. North America	Subarctic (186)	From Yukon River basin, Bonasila, near Fort McPherson, Mackenzie River Delta (AMNH, NMNH, NHM)
	Northwest Coast (39)	Recent inhabitants of Northwest Coast; Kodiak Island, Tlingit, and from Vancouver Islands (NMNH, AMNH, NHM, UC, MH)
	California (118)	Sacramento, Angel Island, San Nicolas, Angeles Bay, Santa Barbara, Santa Rosa, Santa Cruz (NMNH, MH)
	Plateau/Great Basin (46)	Washington, Oregon, Utah, Colorado, Wyoming and Nevada (NMNH, AMNH)
	Arizona (Southwest) (39)	Apache, Sioux, etc. from Arizona (AMNH, NMNH)
	New Mexico (Southwest) (68)	Apache, Pueblo, Navaho, Zuni, etc. (NMNH, AMNH)
	Plain/North (123)	North and South Dakota, Montana, Iowa, Nebraska, Kansas (NMNH)
	Plain/South (94)	Missouri, Arkansas, Texas, Oklahoma (NMNH)
	Northeast Woodland/West (104)	Illinois, Michigan, Wisconsin, Indiana, Ohio (NMNH)
	Northeast Woodland/East (79)	New York, Massachusetts, Connecticut, New Jersey, Pennsylvania, Delaware, Maryland, Kentucky (NMNH)
	Southeast Woodland/North (69)	Virginia, Tennessee, North and South Carolina (NMNH)
	Southeast Woodland/South (80)	Louisiana, Mississippi, Alabama, Georgia, Florida (NMNH)
7. Central/South America	Mexico (131)	Sierra Madre of Durargo, Tombat Xico, Jarasco, Tarasco, Yucatan, Duraugo, Pueblo, Chihuahua, Tarahumare, Papago; Cora, Oroyo del Santa Rosa, Nixtalpa, Territory (AMNH, NMNH,NHM)
	Carib (46)	Recent Native Caribeans from Jamaica, Cuba, Venezuela, Guyana, Guana (NHM, NMNH, AMNH)
	Intermediate (60)	Recent Native Americans from Costa Rica, Panama, Nicaragua, Ecuador, Colombia (NMNH, AMNH, NHM)
	Peru (249)	Cerro del Oro, Cajamarquilla, San Damian, Chilca,Coyugo, Cinco Cerros, Huacho; Casa Grande, Macato, Huaras Lupo, Masca (NHM, NMNH)
	South Andes (37)	Recent Native people from Bolivia, Chile, Argentine (NMH, NMNH, AMNH)
	Fuego/Patagonia (48)	Recent Fuegians from Terra del Fuego and Patagonians (NHM, UC, MH, NMNH, AMNH)
8. India	Nepal (38)	Native of Nepal (NHM)
	Assam/Sikkim (54)	Assam and Sikkim districts (NHM)
	Northeast (164)	Recent inhabitants from West Bengal, Bihar and Orissa (NHM, AMNH)
	Northwest (100)	From Delhi and Punjab district (NHM, AMNH)
	Central (23)	Maharashtra, Nagpur and Bombay (NHM, AMNH)
	Madras (112)	Recent Inhabitants of Madras, Dravidians (NHM)
	South (36)	Recent Inhabitants of Salem, Mysore, and Malabar Coast (NHM, US, AMNH)
	Veddah (27)	Veddah from Ceylon Island, Sri Lanka (NHM, UC)
9. West Asia	Afghanistan (34)	Recent Afghans from Kabul, including several individuals from Peshawar, Pakistan (NHM)
	Iran/Iraq (45)	Dailamanistan, North Iran, Islamic Period (UT) Recent Iraq from Baghdad, Karbala, An Najaf (FM)
	Israel (101)	Recent inhabitants of Israel (NHM, AMNH)
	Syria/Palestine (41)	Bedouin people (AMNH, FM)
	Turkey (56)	Adalia, Aintab, Kurd, Armenian, and Mohammedan mainly from Constantinople (NHM, AMNH)
	Cyprus (32)	Roman and recent periods, Saloi (NHM, AMNH)

(*continued*)

APPENDIX . (Continued)

Large geographic groups	Local groups	Brief information
10. Europe	Russia(46)	Recent Russians (NHM, AMNH)
	Czecho/Poland (53)	Moravia and Bohemia (Czecho), Silesia (Poland) (NHM, SAM, AMNH)
	Hungary (77)	Recent Hungarians from Buda-Pesth, Pressburg, Csakovar, Nagy-Sap, Demko-Hegy (NHM, AMNH)
	Rumania (29)	Recent Rumanians from Transylvania, Ploesci, Wallachia, Bucharest (NHM)
	Yugoslavia (27)	Herzegovina, Dalmatia, Serbia, Montenegro regions, including a few specimens from Bulgaria (NHM, AMNH)
	Greece (40)	From Tripolitza, Morea, Arta, Epirus, Athens (NHM)
	Sweden/Norway (92)	Stockholm, Oslo, Bergen etc. (NHM, UC)
	Holland (31)	Recent Dutch from Groningen, Friesland, Amsterdam, including several individuals from Denmark and Belgium (NHM)
	Germany (50)	Recent Germans from Berlin, Mainz, Holstein (NHM, UC, NMNH)
	Austria/Swiss (67)	Vienna, Tyrol, Modling, Gloggnitz, Malta Carinthia and recent Swiss (AMNH, NMNH)
	Italy (143)	Recent Italians from Napoli, Firenze, Venezia, Milano and other cities (NHM)
	France (74)	Gallo-Roman, Metovingian, and recent French; Recent individuals are mainly soldiers in the Army of Napoleon, including several specimens from Spain and Portugal (NHM)
	United Kingdom Ensay (97)	(Late Medieval to post Medieval period, Scotland)
	Repton (36)	(St. Wystans, Derbyshire, Northeast England)
	Poundbury (114)	(Late Roman period, Southwest England)
	Spitalfields (264)	(Mid-Victorian, Pre-17th century, eastern regions of London) (NHM, UC)
11. North Africa	Pre Dynasty (150)	Badari and Naqada, Egypt (UC)
	Early Dynasty (38)	Lisht, near Matanieh, 12th Dynasty, Egypt (NMNH)
	Middle Dynasty (115)	Cairo, Omdurman, 20th–25th Dynasty, And Gizeh, 26th–30th Dynasty, Egypt (UC, NMNH)
	Recent Egypt (36)	Kharga Oasis, Ain El Turba, El Baguat, Luxor (NMNH)
	Nubia/Dynasty (210)	Kerma, Dinka near Omdurman, 12th–13th Dynasty (UC)
	Nubia/Recent (48)	Hesa, Biga, Sesebi, After Christian date (NHM)
12. Subsaharan Africa	West Africa (85)	Recent West Africans from Gambia, Guinea, Ivory Coast, Liberia, Senegal, and Sierra Leone (NHM, UC, AMNH)
	Ghana/Ashanti (80)	Abuakwa, Dagomba, and Ononguna (NHM, UC, AMNH)
	Nigeria/Ibo (153)	Recent Ibo tribe from Southern Nigeria (NHM, UC)
	Cameroon/Congo (70)	Recent samples from Edia, Ibea, Mandingo, Abuakwa, Duala, Boki, Anyang, Kumabembe, Mabea (Cameroon) Recent populations from Fernand Vaz, Loango, Mobanyki, Selte Cama, Bahuana, Luano (NHM, AMNH)
	Gabon (101)	Ogove River, Fernand Vaz River basin (NHM)
	Somalia (87)	Erigavo District and Darod Kuhar, Somalia, including a few specimens from recent Ethiopians (NHM, AMNH)
	Kenya (160)	Recent inhabitants of Kikuyu, Nairobi, Teita Hills (UC, AMNH)
	Tanzania (151)	Recent Haya tribe from Lake Victoria, Pare, Gonja, Angoni (NHM, UC, AMNH)
	Uganda (19)	Recent Teso tribe, Ankokle, Wakmia, Chif Ruakirara, including several individuals from Rwanda (Mtussi, Mhutu) (NHM, AMNH)
	South Africa (27)	Recent inhabitants from Zambia, Zimbabwe, Malawi, Mozambique, Lesotho (NHM, UC, AMNH)
	Zulu (100)	Recent Zulu from Pietremanitzburg, Mantatee, Tulu, Tambuki, Natal, South Africa (NHM, UC, AMNH, NMNH)
	Khoi-San (49)	Recent Bushmen and Hottentots from South Africa (NHM, UC, AMNH)

UT, University of Tokyo, Tokyo, Japan; KU, Kyoto University, Kyoto, Japan; NHM, Natural History Museum, London, UK; UC, University of Cambridge, Cambridge, UK; MH, Musee del'Homme, Paris, France; AM, Australian Museum, Sydney, Australia; US, University of Sydney, Sydney, Australia; SAM, South Australian Museum, Adelaide, Australia; BM, Bishop Museum, Honolulu, USA; NMNH, National Museum of Natural History, Smithsonian Institution, Washington D.C., USA; AMNH, American Museum of Natural History, New York, USA; FM, Field Museum, Chicago, USA.

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